

Toronto 311 Service Request Analytics

Business Intelligence & Operational Performance Report

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Business Intelligence | Data Analytics | Tableau | SQL | Python

Executive Summary

The City of Toronto manages a complex municipal service delivery ecosystem responsible for responding to thousands of resident service requests across transportation, infrastructure, public works, waste management, municipal licensing, and environmental services. As service demand continues to increase alongside urban population growth, municipal leaders require timely and accurate operational intelligence to ensure efficient allocation of public resources and maintain high-quality service delivery.

This Business Intelligence project presents a comprehensive analysis of Toronto's 311 Service Request data collected between January and May 2026. Using Tableau, interactive dashboards were developed to transform operational records into executive-level reporting tools capable of supporting strategic planning, operational monitoring, and evidence-based decision-making.

Three integrated dashboards were designed to examine municipal operations from complementary perspectives. The Executive Service Request Overview provides a high-level summary of service demand, operational status, and monthly performance trends. The Division Performance Dashboard evaluates workload distribution across municipal departments, enabling comparison of operational activity and resource utilization. Finally, the Geographic Analysis Dashboard visualizes service request density across Toronto wards using spatial analysis techniques, allowing stakeholders to identify geographic hotspots and areas requiring additional operational attention.

The analysis identified clear differences in service demand across wards, with higher request concentrations observed in densely populated urban areas. Operational trends demonstrated that transportation and infrastructure-related requests accounted for a significant proportion of municipal workload. Request completion metrics indicated generally strong operational performance while also highlighting opportunities for continuous process improvement.

Beyond descriptive reporting, the dashboards provide an interactive decision-support platform capable of assisting municipal managers in workforce planning, budget allocation, service prioritization, and operational performance monitoring. Geographic visualization further enhances situational awareness by enabling rapid identification of service demand patterns across the city.

The project demonstrates how modern Business Intelligence techniques can transform municipal operational data into meaningful, actionable insights. By integrating interactive dashboards, spatial analytics, calculated performance indicators, and executive reporting principles, the solution establishes a scalable analytical framework capable of supporting future predictive analytics, service forecasting, and smart city initiatives.

Dashboard 1 Analysis

Executive Operational Performance

Overview

The Executive Dashboard provides a consolidated operational overview of Toronto's 311 service requests during the January–May 2026 reporting period. The dashboard is designed for municipal executives who require rapid access to key operational metrics without navigating detailed reports.

A total of **237,681** service requests were recorded across seven municipal divisions. Overall operational performance remained strong, with **75.8%** of requests successfully completed during the reporting period. Transportation Services managed the largest operational workload, accounting for **105,051** service requests, approximately **44%** of all requests, highlighting the significant demand placed on transportation infrastructure.

Service Requests by Division

Transportation Services received more than **105,000** requests, making it the largest contributor to municipal service demand. Solid Waste Management followed with **56,774** requests, while Municipal Licensing & Standards processed **42,109** requests. Together, these three divisions accounted for the majority of all municipal service requests, suggesting that transportation infrastructure, waste collection, and municipal regulation remain the city's most resource-intensive operational functions. Lower request volumes were observed within Parks and Recreation (**2,673**) and Environmental Services (**9,893**), indicating comparatively lower public demand during the study period.

Business Insight

The concentration of requests within Transportation Services suggests continued investment in transportation infrastructure maintenance, road repair, and winter operations should remain a strategic priority.

Request Status Distribution

The dashboard indicates that **180,221** requests were successfully completed, representing approximately **75.8%** of all submitted requests. Smaller proportions remained In Progress (**21,645**), Cancelled (**20,086**), Closed (**3,293**), New (**9,494**), or Unknown (**2,942**).

Business Insight

A completion rate exceeding 75% demonstrates strong operational performance; however, the presence of more than **21,000** in-progress requests highlights opportunities to improve service turnaround times through workflow optimization and resource balancing.

Top Service Request Types

Road maintenance requests including road pothole and road damage reports—represented the highest-volume service category. Sidewalk snow clearing and road plowing requests also ranked among the most frequent municipal services.

Business Insight

The prominence of transportation-related requests reinforces the operational importance of roadway maintenance and seasonal infrastructure management within Toronto's municipal operations.

Monthly Trend Analysis

Overall service requests declined steadily between January and March before showing moderate recovery in April and May. This trend may reflect seasonal demand patterns, operational scheduling, or changing environmental conditions.

Business Insight

Municipal managers should anticipate higher operational workloads during early winter and early spring periods and adjust staffing levels accordingly.

Dashboard 2 – Division Performance Analysis

Division Performance Dashboard

Purpose

The Division Performance Dashboard provides a comparative analysis of operational activity across Toronto's municipal divisions. While the Executive Dashboard presents an organization-wide summary, this dashboard focuses on departmental performance by examining workload distribution, request status, monthly demand trends, and geographic service concentration. The dashboard enables operational managers to answer several key questions:

- Which municipal divisions receive the highest service demand?
- How has demand changed over time?
- Which wards generate the greatest workload?
- How is work progressing across different operational units?
- Where should staffing and operational resources be prioritized?

By integrating workload, status, and temporal analysis into a single interface, the dashboard supports evidence-based operational planning and performance monitoring.

Division Performance by Status

The stacked bar chart compares service request volumes across Toronto's municipal divisions while simultaneously illustrating the distribution of request statuses within each division. Transportation Services processed the largest operational workload by a significant margin, with over **100,000 service requests** during the reporting period. The overwhelming majority of these requests were completed successfully, while a smaller proportion remained in progress or were cancelled. Solid Waste Management ranked as the second busiest division, followed by Municipal Licensing & Standards. Toronto Water, Environmental Services, Parks and Recreation, and the 311 Call Centre experienced considerably lower service volumes. The visualization demonstrates that although workload differs substantially across divisions, completed requests consistently represent the largest operational outcome.

Business Insight

The concentration of completed requests across all divisions indicates consistent operational execution despite significant differences in workload. Transportation Services carries a disproportionately high operational burden, suggesting that staffing levels, maintenance budgets, and infrastructure investments should continue to prioritize transportation-related services.

Monthly Division Trend Analysis

The monthly trend chart illustrates how service demand changed between January and May 2026 for each municipal division.

Several important operational patterns emerge:

- Transportation Services experienced the highest request volume throughout the reporting period, beginning with approximately **33,341 requests** in January before declining to **13,424 requests** by May.
- Solid Waste Management experienced a decline during February before recovering steadily throughout the remaining months.
- Municipal Licensing & Standards showed gradual growth after March, suggesting increasing operational activity later in the reporting period.
- Toronto Water maintained relatively stable demand with only modest fluctuations.
- Parks and Recreation and Environmental Services remained comparatively stable with consistently lower request volumes.

Overall, municipal service demand declined after January before stabilizing during April and May, indicating potential seasonal influences on service activity.

Business Insight

The sharp reduction in Transportation Services requests between January and March likely reflects seasonal operational cycles. Winter maintenance activities—including snow clearing, road treatment, and weather-related infrastructure requests—typically generate elevated service demand during the early months of the year. As weather conditions improve, operational demand naturally decreases before stabilizing. This seasonal pattern highlights the importance of flexible workforce planning and adaptive resource allocation to accommodate fluctuations in service demand.

Rank	Ward	Requests
1	Toronto–Danforth (14)	14,920
2	Davenport (09)	14,213
3	Beaches–East York (19)	12,822
4	University–Rosedale (11)	12,698
5	Eglinton–Lawrence (08)	12,263
6	Toronto–St. Paul's (12)	11,984
7	Parkdale–High Park (04)	11,888
8	Etobicoke–Lakeshore (03)	11,760
9	Scarborough Southwest	10,441
10	York South–Weston (05)	10,424

The relatively narrow gap between the top-ranked wards indicates that service demand is distributed across several densely populated urban areas rather than being concentrated in a single location.

Business Insight

High-demand wards should be monitored continuously to ensure that municipal resources remain aligned with resident needs. Geographic demand patterns provide valuable information for

infrastructure investment, staffing decisions, preventative maintenance programs, and service planning.

Operational Findings

The Division Performance Dashboard reveals several important operational characteristics of Toronto's municipal service delivery system:

- Transportation Services consistently experiences the highest operational workload across every month analyzed.
- Service demand exhibits clear seasonal fluctuations, particularly within transportation-related operations.
- Completed requests dominate across all municipal divisions, reflecting effective service delivery.
- Service demand is concentrated within a relatively small number of high-density wards.
- Municipal operations appear well balanced across departments despite substantial differences in workload.

RECOMMENDATIONS

- Increase staffing and equipment availability during periods of historically high transportation demand, particularly during winter months.
- Develop ward-specific operational plans for Toronto–Danforth, Davenport, Beaches–East York, and other consistently high-demand areas to reduce response times.
- Establish monthly KPI reviews for each municipal division to identify emerging workload imbalances and improve operational efficiency.
- Leverage historical service request data to forecast future demand and proactively allocate municipal resources before seasonal peaks occur.

Observation: This dashboard demonstrates more than descriptive reporting. it supports operational management. By combining departmental performance, temporal trends, and geographic demand into a single analytical view, it enables managers to identify workload patterns, evaluate service delivery, and make informed resource allocation decisions.

Dashboard3 - Geographic Distribution Dashboard

Purpose

The Geographic Distribution Dashboard provides a spatial analysis of Toronto's 311 service requests at the ward level between January and May 2026. While the previous dashboards focused on organizational performance and operational trends, this dashboard answers a critical question for municipal decision-makers:

Where are service requests occurring?

By integrating Toronto's ward boundary GeoJSON with operational service request data, the dashboard enables city managers to identify geographic demand hotspots, compare service activity across wards, and support location-based planning.

Unlike traditional tabular reports, spatial visualization allows decision-makers to immediately recognize regional differences in municipal service demand, making it easier to prioritize infrastructure investment, workforce deployment, and community service planning.

Geographic Distribution of Service Requests

The choropleth map presents the total number of service requests submitted within each Toronto ward. Darker shades represent higher service request volumes, while lighter shades indicate comparatively lower demand. The visualization reveals a clear spatial variation in municipal service activity across the city. Central and southern wards generally display higher concentrations of requests, while several northern and peripheral wards experience lower request volumes. Rather than being evenly distributed, service demand is concentrated within specific geographic areas, reflecting differences in population density, land use, transportation infrastructure, commercial activity, and community development. The interactive tooltip allows users to examine each ward individually, displaying:

- Ward name
- Total service requests
- Completion rate

This functionality provides both a city-wide overview and detailed ward-level performance analysis.

Business Insight

Spatial clustering of service requests suggests that municipal demand is heavily influenced by urban characteristics. High-density residential neighbourhoods, major transportation corridors, and commercial districts naturally generate greater service demand, requiring targeted operational planning.

Top Service Request Types

Transportation-related requests dominate municipal operations, with road maintenance and winter service activities representing the highest demand categories. The dashboard confirms that road pothole repairs, sidewalk snow clearing, and road plowing requests continue to generate significant operational workloads.

Business Insight

The concentration of transportation-related requests reinforces the strategic importance of maintaining roadway infrastructure and winter maintenance programs, particularly within high-demand wards.

Monthly Geographic Trend

The monthly trend indicates that citywide service requests declined after January before stabilizing during April and May. Although overall demand decreased, the geographic distribution of requests remained relatively consistent throughout the reporting period, suggesting that high-demand wards continue to require sustained operational attention regardless of seasonal fluctuations.

Business Insight

Historical geographic trends provide valuable information for forecasting future service demand and developing long-term municipal resource allocation strategies.

Spatial Analysis Findings

The Geographic Distribution Dashboard reveals several important characteristics of Toronto's municipal operations:

- Service requests are not evenly distributed across Toronto.
- Urban wards consistently generate higher municipal service demand.
- Transportation-related issues remain the dominant operational concern across multiple geographic regions.
- Strong completion rates indicate effective service delivery despite varying workload levels.
- Geographic visualization enables rapid identification of operational hotspots that would be difficult to recognize using tables alone.

Observation: The Geographic Distribution Dashboard demonstrates the value of combining **Business Intelligence** with **spatial analytics** to support public sector decision-making. Rather than presenting operational data solely through traditional charts, the dashboard enables stakeholders to understand **where** municipal services are most needed, providing a powerful foundation for strategic planning, infrastructure investment, and equitable resource allocation. By integrating interactive mapping, ward-level performance metrics, service categories, and operational KPIs, this dashboard elevates the analysis from descriptive reporting to actionable

decision support. It illustrates how geographic intelligence can strengthen municipal operations and improve service delivery for residents.